

**Daniel RACOCEANU**

Professor

Sorbonne UniversityFaculty of Science and Engineering, Eng. Dpt. &
Faculty of Medicine

☎ +33 (0)6 33 77 42 67

✉ daniel.racoceanu@sorbonne-universite.fr

🌐 sorbonne-universite.fr/en/university

**Paris Brain Institute / Institut du Cerveau**

CNRS UMR 7225 – Inserm U 1127 – Sorbonne University UM75

ARAMIS Lab – *Inria* • Office : 3.008

Hôpital Pitié-Salpêtrière, 47 boulevard de l'Hôpital, 75013 Paris, France

☎ +33 (0)1 57 27 47 71 • ✉ daniel.racoceanu@icm-institute.org

🌐 <https://institutducerveau-icm.org/en/>**Paris Brain
Institute****Inserm****BIOGRAPHY**

I am a University Professor specializing in Computational Pathology, Biomedical Imaging, Pattern Recognition, Machine and Deep Learning, and Explainable/Responsible AI at Sorbonne University, and a Principal Investigator at the Paris Brain Institute (ICM) within the Aramis Lab. My research focuses on computational pathology, spatial transcriptomics, and microscopic biomedical image analysis, with a particular emphasis on developing novel, explainable, and trustworthy AI paradigms for biomedical applications.

I earned my Habilitation (HDR) in 2006 and my Ph.D. in 1997 from Marie and Louis Pasteur University (formerly the University of Franche-Comté, France). Prior to entering academia, I worked as a Project Manager at General Electric. In 1999, I joined Marie and Louis Pasteur University as an Associate Professor and Research Fellow at the FEMTO-ST Institute (CNRS UMR 6174).

Between 2005 and 2014, I contributed to the creation and development of the International Joint Research Unit IPAL (UMI CNRS 2955) in Singapore, serving as its Director from 2008 to 2014. IPAL was established through a partnership between Sorbonne University, CNRS, the National University of Singapore (NUS), and A*STAR. During this period (2009–2015), I also held a Professorship at the School of Computing, National University of Singapore.

From 2014 to 2016, I was a member of the Executive Board of the University Institute of Health Engineering at Sorbonne University and Co-Director of the Cancer Theranostics research team at the Bioimaging Laboratory (LIB – CNRS UMR 7371, Inserm U1146).

Between 2016 and 2018, I served as a Professor of Biomedical Imaging and Data Analytics at the Pontifical Catholic University of Peru.

I have been actively involved in international scientific governance. I served as Vice-President (2016–2018) and President (2018–2020) of the European Society for Digital and Integrative Pathology (ESDIP) and have been a member of its Advisory Board since 2020.

From 2018 to 2022, I was a Member of the Board of Directors of MICCAI (Medical Image Computing and Computer-Assisted Intervention), serving as General Chair of MICCAI 2020 and contributing to the organization of MICCAI 2021 and 2022.

PROFESSIONAL & RESEARCH EXPERIENCE

Since May 2019 – **Sorbonne University, Paris Brain Institute (ICM)**, Paris, France

- **Principal Investigator, Paris Brain Institute (Institut du Cerveau – ICM)** - Inserm, CNRS, AP-HP
- **Principal Investigator, INRIA team-project Aramis** (National Institute for Research in Digital Science and Technology).
- **Full Professor @ Faculty of Science & Engineering, Engineering & Computer Science Department**
 - Courses & Labs:
 - Artificial Intelligence, Machine / Deep Learning (master – Eng.) – coordinator
 - Medical and Biomedical Image Analysis (master level – Eng.) – coordinator
 - Biomedical Image Analysis (master level – Eng.) – coordinator
 - Computer Graphics (master level – Computer Sciences) – coordinator
 - Object Oriented Programming (Master level – Eng.)
 - Programming (undergraduate level – all options)
- 2018-2022 – Member of the **Board of Directors** of the international **MICCAI Society**
- 2020 – general Chair MICCAI 2020¹ (Medical Image Computing and Computer Assisted Intervention)
- Since 2020 – **Advisory Board** member - **ESDIP**² (European Society of Digital Integrative Pathology)
- 2018-2020 – **President of ESDIP** – the European Society of Digital Integrative Pathology

2018 – 2019: Kitview, Orqual group, Lyon, France

- **Scientific Director:**
- Installing the R&D HQ @ Insavalor Lyon (INSA Lyon's industrial ecosystem).
- Building a local R&D team.
- Setting up a CIFRE PhD in collaboration with LIRMM CNRS lab, Montpellier, France.
- Initiating an R&D collaboration (Spin-Off) in Cluj-Napoca, Romania.

2016 – 2018: **Pontifical Catholic University of Peru (PUCP)**, San Miguel, Lima, Peru

- **Full Professor @ Faculty of Science and Engineering, Department of Engineering;**
- Master and PhD students' supervision @ the Medical Imaging Lab;
- European & international initiatives around Computational and Integrative Pathology;
- Succeeding to the **MICCAI 2020** bid - Medical Image Computing & Computer Assisted Intervention (MICCAI 2020 conference was hosted for the first time by Latin America);
- 2018 – Member of **MICCAI Society Board of Directors** (Medical Image Computing and Computer Assisted Intervention)

2011- 2016: **University Pierre and Marie Curie**, Paris, France

- **Full Professor @ Engineering Department.**
- 2014-2016 - **Research Director** - cancer diagnosis and therapies team, **Biomedical Image Lab** (LIB³ - CNRS UMR 7371, Inserm U1146);
- 2008-2014 – **Director** of the **CNRS** research lab **IPAL** (UMI CNRS 2955) **Singapore**
- **Executive Board, University Institute of Health Engineering (IUIS) of Sorbonne University**
- Co-Director of a new Bachelor of Sciences minor: "Innovation in Public Health"
- 2016 – co-founder of the **ESDIP** (European Society of Digital Integrative Pathology)
- 2016-2018 – Vice-president of the **ESDIP** (Eur. Society of Digital Integrative Pathology).

2013 – 2015: **National University of Singapore (NUS)**, Singapore

- **Full Professor** (adjunct) @ School of Computing, Computer Science Dpt.
- 2008-2014 – **Director** of the **CNRS** research lab **IPAL - UMI CNRS 2955 Singapore**
- NUS SoC PhD co-supervision; NUS SoC PhD reviews and jury member.

2011 – 2014: **French National Center for Scientific Research (CNRS)**

- **CNRS Research Director in Biomedical Engineering and Imaging**
- Second mandate as **IPAL UMI CNRS 2955 Director**.

2008 – 2014: **Director, International Joint Research Lab IPAL UMI CNRS**, Singapore

- Restructuration around two impactful areas: biomedical image understanding and ambient assisted living; successful CNRS et al. evaluations in 2009 and 2014; growth of the lab by

¹ MICCAI 2020: <https://www.miccai2020.org/en/ORGANIZING-COMMITTEE.html>

² ESDIP (European Society of Digital Integrative Pathology) : <https://www.esdipath.org/>

³ Lab. Imagerie Biomédicale, LIB - UMR CNRS UMR7371, INSERM U1146: <https://www.lib.upmc.fr/about-lib/?lang=en>

bringing in the University Pierre and Marie Curie and Mines-Télécom Institute.

2009 – 2013: **National University of Singapore (NUS)**, Singapore

- **Associate Professor** (adjunct) @ **School of Computing**, Computer Science Dpt.
- NUS SoC PhD students' co-supervision, PhD reviews and PhD juries
- Co-PI of the A*STAR SERC project "MMedWeb"
- Collaborator in a 2011 Singapore - MIT Alliance (SMA3)

2005 – 2011: **French National Center for Scientific Research (CNRS)**

- **CNRS Research Fellow (Chargé de Recherche CNRS)**, creation and management of the Medical Image Indexing team at the International Research Unit PAL (UMI CNRS).
- Research focusing on Medical Image Analysis, Content Based Image Retrieval, Medical Ontologies, Medical Multimedia Fusion, Cognitive Vision.
- Participation @ CLEF international benchmark, Medical Image Indexing-Retrieval track.

1999 – 2008: **Marie and Louis Pasteur University** (former Franche-Comté Univ), Besançon, France

- **Associate Professor**, Faculty of Sciences & Technologies, Sciences & Technologies Dpt.
- Courses at Sciences and Techniques Dpt. in artificial intelligence, computer vision, diagnosis, prognosis, control sciences, fault detection, real-time, non-linear, and discrete event systems.
- Involved in the creation of the new LMD (Licence-Master-Doctorat) program.
- Research Fellow at **FEMTO-ST Institute**⁴ (UMR CNRS 6174), Besançon, France, in AI applied to dynamic monitoring, diagnosis, and prognosis, in the e-maintenance framework.

1998 – 1999: **General Electric Power Generation**⁵, Belfort, France (€1,2 B; 2000 pers.)

- **Project Manager**, General Electric Gas turbine department
- Design, technical, production, logistics, financial management gas turbines projects
- Reengineering, reorganization for General Electric group integration teams.
- Optimization of the management flow.
- Detection of financial niches and consolidation actions in the financial flow.
- Design of common project management and client management protocols between Alstom (old owner of the company) and General Electric (new owner of the company)

1997 – 1998: **Gaussin Manugistique**⁶, Héricourt, France, (M€10 /100 persons)

- Design & manufacturing of special trailers for industry, airports, and seaports.
- **Logistics & Planning Manager** - Logistics management, supply chain management
- Reengineering and management of a distributed subsidiary structure.

1993 – 1996: **University of Franche-Comté**, Univ. Institute Technology Belfort, France

- **Lecturer** (ATER - Attaché Temporaire d'Enseignement et de Recherche)
- Dpt. Production System Management, University Institute of Technology of Belfort, University of Franche-Comté, Belfort, France
- Logistic management, supply chain management courses and laboratory
- Flexible production system labs (RFID, PLC programming, production flow management); I created a set of hands-on laboratories on a state-of-the-art flexible industrial production system.

1992: **Research Institute for Welding and Material Testing**⁷ (ISIM), Timisoara, Romania

- **Research Fellow** in welding robotics.

PROFESSIONAL ORGANIZATIONS

Medical Image Computing and Computer Assisted Intervention (MICCAI Society)

- 2018-2022: member of the Board of Directors and General Chair of MICCAI 20204

European Society of Digital and Integrative Pathology (ESDIP3)

- Since 2020 - ESDIP Advisory Board
- 2018 to 2020 - President & Member of the Executive Committee
- 2016 to 2018 - Vice-President & Member of the Executive Committee

⁴ FEMTO-ST Institute (UMR CNRS 6174), Besançon, France: <https://www.femto-st.fr/en>

⁵ General Electric Power Generation: <https://www.gevernova.com/regions/europe>

⁶ Gaussin Manugistique: <http://www.gaussin.com>

⁷ National R&D Institute for Welding and Material Testing (ISIM), Timisoara, Romania: <https://www.isim.ro/en/>

French Chamber of Commerce from Singapore (FCCS)

- 2011 to 2014 – co-President & co-founder of the R&D Committee of the FCCS.

MOBILITY ACROSS THEMATIC AREAS

Since 2016 Integrative Digital Pathology & XAI - Prof. @ Sorbonne University and Prof. @ PUCP
2012-2016 Computational Pathology – Prof. @ UPMC & Prof. @ NUS & DR @ CNRS
2008-2012 BioMedical Imaging, Cognitive BioMedical Image, Stochastic Prior shape and Simplicial models – Prof. @ UPMC, Prof. @ NUS, DR & CR @ CNRS
2005-2008 Content-Based Medical Image Indexing and Retrieval reinforced by Semantic Approaches - A/Prof. @ UFC, A/Prof. @ NUS, CR @ CNRS
1999-2005 Dynamic Pattern Recognition for Diagnosis/Prognosis using dynamic recurrent neural networks - A/Prof. @ **Marie and Louis Pasteur University** (univ. Franche-Comté)
1992-1997 Stochastic Models Simplification and Control using Principal Component Analysis, PhD candidate @ UFC, Lecturer @ Univ. Inst. Techn. Belfort, M.Sc. @ UTBM
1987-1992 Mechatronics - M.Eng. @ Politehnica University of Timisoara (UPT)

GEOGRAPHIC MOBILITY

Since 2019 Paris, France - Prof. @ Sorbonne University, PI @ Paris Brain Institute (ICM)
2018-2019 Lyon, France - Scientific Director @ Kitview, Orqual Group
2016-2018 San Miguel, Lima, Peru - Prof. @ Pontifical Catholic University of Peru (PUCP)
2014-2016 Paris, France - Prof. @ University Pierre and Marie Curie (UPMC Univ Paris 6)
2005-2014 Singapore - CNRS Senior Research Fellow & UMI CNRS Lab Director
1999-2005 Besançon, France - A/Prof. @ Marie and Louis Pasteur University (U. Franche-Comté)
1998-1999 Belfort, France - Project Manager @ GE Energy Europe Technol. Center
1997-1998 Héricourt, France - Logistic & Planning Manager @ Gaussin S.A.
1993-1997 Belfort, France – PhD candidate @ University of Franche-Comté & Lecturer (ATER) @ Belfort-Montbéliard University Institute of Technology (IUT)
1992-1993 Belfort, France - Master of Sciences @ Univ. of Technology Belfort-Montbéliard (UTBM)
1987-1992 Timisoara, Romania - Master of Engineering @ Politehnica Univ. Timisoara (UPT)

EDUCATION

2006 **HDR (Accreditation to Supervise Research / Habilitation à Diriger des Recherches)**
- Control and Computer Science
University of Franche-Comté, Besançon, France
- Dynamic Monitoring using Artificial Intelligence Techniques
- **Keywords:** Dynamic Monitoring, Artificial Intelligence, Dynamic Neural Networks, Neuro-Fuzzy Systems, Fuzzy Petri Nets, Diagnosis, Prognosis.

1997 **Ph.D. Control and Computer Sciences, University of Franche-Comté, Besançon, France**
- Stochastic model reduction. Quasi-optimal management solution of an EDF (Electricity of France) Hydropower System (a system with 7 centrals).
- **Keywords:** Stochastic Modelling, Markov Chains, Control, Reduction Methods, Singular Perturbations - Principal Component Analysis.
(*Highest distinction - jury's congratulations – très honorable avec félicitations du jury*).

1993 **M.Sc. (Master of Science) Control Sciences**
University of Technology of Belfort-Montbéliard, France
(*Major & with Distinction – Très Bien*).

1992 **Dipl. Ing. (Master of Engineering) - Mechanical Manufacturing Engineering Technology**
Politehnica University of Timisoara, Romania
(*Highest Distinction in Romanian academic system – 10/10*).

CURRENT SCIENTIFIC ACTIVITY

General interest: Responsible Artificial Intelligence, High-content Biomedical Image analysis
Responsible Artificial Intelligence, Explainable AI, Computer science for biomedical data, information and knowledge analysis / management. Integrative biomedical image analysis.

Specific area of expertise: Explainable Artificial Intelligence / Deep Learning, Integrative approaches for biomedical data analysis, Semantic-driven high-content image exploration

Explainable Artificial Intelligence / Deep Learning, Integrating heterogeneous models for omics and high-content imaging data understanding, with initial (not exclusive) focus on digital pathology. Semantic-driven, traceable deep-learning and scalable image analysis.

Keywords: High-content biomedical image analysis, integrative digital pathology, semantic-driven high-content biomedical image analysis, explainable and responsible deep learning / artificial intelligence, stochastic modelling and prior shapes, mathematical morphology on sparse-sets, content-based image retrieval.

SCIENTIFIC PUBLICATIONS

Exhaustive publication list: <https://daniraco.github.io/publications.html>

Google Scholar: <https://scholar.google.com/citations?user=2eBRLj0AAAAAJ&hl=fr>

Research Gate: https://www.researchgate.net/profile/Daniel_Racoceanu

ORCID: <http://orcid.org/0000-0002-9416-1803>

Web of Science Researcher-ID: F-4576-2011

DBLP profile: <https://dblp.org/pid/63/4416.html>

Semantic Scholar ID: <https://www.semanticscholar.org/author/Daniel-Racoceanu/1742773>

SCIENTIFIC SUPERVISION

- **Co-guarantor Dr. Habil – habilitation to manage scientific research (HDR)**
 - Dr Nicolas LOMENIE, A/Prof, Univ Paris Descartes, IPAL Res. Fellow Jan.2009-Sep.2012
- **Post-docs supervised**
 - Dr Anuradha KAR (ICM, 2021- 2023) - Alzheimer's disease patients' stratification
 - Dr Janan ARSLAN (ICM, 2021-2023)- model. metabolic plasticity heterogeneity in melanoma
 - Dr Antoine VEILLARD (UPMC, 2013-2016) - collaborative digital pathology
 - Dr Ludovic ROUX (UJF, 2008-2014) - semantic cognitive approaches
 - Dr HUANG Chao-Hui (NUS and A*STAR, 2008-2014) - cell tracking and WSI analysis.
 - Dr Maria KULIKOVA (CNRS/UPMC 2010-2013) - marked point process, nuclei detection.
 - Dr Caroline LACOSTE (CNRS, 2005-2006) – marked point process angiography analysis
- **On-going and graduated PhDs:**
- *On-going PhD supervision:*
 - M. Mehdi HAMADACHE (start Oct. 2025, Sorbonne Univ., 100%) – spatial transcriptomics using physically inspired AI.
 - Ms. Swann RUYTER (start Oct. 2024, Sorbonne Univ., 100%) – virtual staining, virtual proteomics and virtual spatial transcriptomics using generative IA.
 - Ms. Esther KOZLOWSKI (start Oct. 2023, Sorbonne Université, 50%) – analysis of propagation pathways in Parkinson's disease using explainable AI.
 - Ms. Ayse GUNGOR (start Aug. 2023, Sorbonne Université, 50%) – eye and brain.
 - M. Ilias SARBOUT (start Oct. 2023, Sorbonne Université, 50%) – artificial vision.
- *Graduated PhDs supervised / first position after PhD:* <http://www.theses.fr/fr/?q=racoceanu>
 - Ms. Laura MARIN (Feb. 2025, PUCP, Lima, Pérou) – Digital Integrative Pathology for personalized medicine, R&D engineer - AREAS Grenoble, France.
 - M. Mehdi OUNISSI, (Oct. 2024, Sorbonne Univ., Paris) - Explainable Artificial Intelligence, post-doc researcher at the Rothschild Foundation, Paris, France.
 - M. Gabriel JIMENEZ, (Sept. 2024, Sorbonne Univ., Paris) - Interpretable Deep Learning in Computational Histopathology for Alzheimer Disease Patients' Stratification Refinement, Research engineer at Paris Brain Institute, Paris, France.
 - Ms. Oumeima LAIFA, (Sept. 2019, Sorbonne Univ., Paris) - A Joint Discriminative-Generative Approach for Tumor Angiogenesis Assessment Comp. Pathology, R.Eng.Owkin.
 - M. Lamine TRAORE, (Dec. 2017, UPMC, Paris) - Semantic Modelling of a Histopathology Image Exploration & Analysis Tool, CEO smart'GRAD – UPMC spin-off.
 - M. Bassem BEN CHEIKH (Sept. 26, 2017, UPMC, Paris) - Graph-based Mathematical Morphology for Characterizing the Spatial Organization of Histological Structures in High-Content Images: Appl. Tumor Microenvironment in Breast Cancer.
 - M. Olivier MORERE (June 2016, UPMC, Paris) - Deep Learning Compact and Invariant

Image Descriptors for Instance Retrieval.

- Ms. Sreetama BASU (March 2015, NUS, Singapore) - Digital Reconstruction of Neuronal Structures from 3D Microscopy data / Research Fellow @ École Normale Supérieure (Ulm) Paris, Institute of Biology, Bioinformatics & Comput. Biology group;
- M. Antoine FAGETTE (June 2014, UPMC, Paris) - Dense Crowd Analysis / Research Engineer THALES Singapore;
- M. Stéphane RIGAUD (March 2014, UPMC, Paris) - Analysis-Synthesis Approach for Neurosphere Modelisation Under Phase-Contrast Microscopy / 3 years Research Fellow @ Institut Curie, Paris;
- M. Humayun IRSHAD (Jan. 2014, UJF, Grenoble) - Automated Mitosis Detection in Color and Multispectral High-Content Images in Histopathology: Appl. to Breast Cancer Grading in Digital Pathology / Res. Fellow (3 years contract) @ Harvard Medical School, Boston, US
- M. Antoine VEILLARD (Dec. 2012, NUS, Singapore) - Kernel Methods for the Incorporation of Prior-Knowledge into Support Vector Machines. Application to whole slide image analysis for breast cancer grading / Research Fellow (3 years) @ Univ. Pierre & Marie Curie, Paris;
- Ms. Roxana TEODORESCU (Apr. 2011, Univ. Besançon) - Parkinson's Disease Prognosis using Diffusion Tensor Imaging Features Fusion / Research Fellow (3 years) @ Mount Sinai Hospital, New York, USA.
- Ms. Adina TUTAC (Oct. 2010, Univ. Besançon) - Formal Representation and Reasoning for Microscopic Medical Image-Based Prognosis. Application to Breast Cancer Grading / R&D Manager @ Sionic SRL, Timisoara, Romania.
- M. Nicolas PALLUAT (Jan. 2006, Univ. Besançon) - Dynamic monitoring using temporal neuro-fuzzy syst. / Res. Fellow (3 years contract) @ Univ. Federal Santa Catarina, Brazil;
- Ms. Eugenia MICA (Sept. 2004, Univ. Besançon) - Discrete events systems monitoring using fuzzy Petri nets / A/Prof @ Univ. Valahia, Târgoviste, Romania.
- M. Ryad ZEMOURI (Nov. 2003, Univ. Besançon) - Monitoring using dynamic neural networks A/Prof. CNAM, Paris, France.

SUCCESSFUL FUNDRAISING

Major competitive projects:

- **Interdisciplinary approaches in oncogenic processes and therapeutic perspectives: Contributions of mathematics and informatics to oncology – MIC 2025 - Synergizing Mechanistic and AI Approaches for Modeling the Impact of Microenvironment Heterogeneity on Immunotherapy efficacy (SIMAI)**
 - Partners: Paris Brain Institute (CNRS UMR7225 – Inserm U1127), the University of Montpellier (LPHI - CNRS UMR5235), and the Montpellier Cancer Research Institute - IRCM (Inserm U1194).
 - Role: PI. Personal contribution: multimodal biomedical imaging analysis using frugal AI
 - Duration: 4 years (Oct. 2025 – Sept. 2029). Budget: 528 400 €.
- **Big Brain Theory (BBT3), Paris Brain Institute - STRATIFIAD: Refining Alzheimer Disease Patients' stratification using effective, traceable and explicable artificial intelligence approaches in computational histopathology.**
 - Leader of the project. Personal contribution: WSI analysis and XAI / DL approaches
 - Partners: INRIA team Aramis lab, Alzheimer team, and Data analysis center, Paris Brain Institute
 - Duration: 2,5 years (July 2021 - December 2023). Budget 200 000 €.
- **AVIESAN (French Alliance for Life and Health Sciences) - ITMO Cancer - Mathematical Approaches to Modelling Metabolic Plasticity and Heterogeneity in Melanoma (MALMO)**
 - Partners: Paris Brain Institute (CNRS UMR7225, Inserm U1127), University of Montpellier (LPHI CNRS UMR5235, LIRMM (CNRS UMR 5506) and Montpellier Cancer Research Institute - IRCM (Inserm U1194)
 - Role: PI. Personal contrib.: traceable DL for 3D histological reconstruction and tumor characterization
 - Duration: 3 years (Nov. 2020 – Oct. 2023). Budget: 400 000 €
- **EU EIT Health: PAPHOS - Platform for advanced prescriptive health operational system**
 - Granted by the European Institute of Innovation & Technology (EIT) - EIT Health
 - Big data in healthcare (digital pathology, sleep apnea)
 - URL - PAPHOS project: <https://www.eithealth.eu/paphos>
 - URL - EIT Health: <http://eit.europa.eu/eit-community/eit-health>
 - Consortium: ATHOS Spain SAE, BULL SAS, CEA, Univ. Pierre et Marie Curie, Univ. Joseph Fourier (UJF), Univ. Politécnica de Madrid, GMV, Karolinska Institutet (KI), AVENTYN.
 - Role: PI. Personal contribution: Digital Pathology use-case with Pitié-Salpêtrière Hosp., Paris.

- Duration: 2 years (Jan. 2016 - Dec. 2018). Budget: 1.400.000 €.
- **Medical Research Foundation (FRM, France); Bioengineering for health**
 - **Multiparametric ultrasonic classification to evaluate tumor progression**
 - Partners: Sorbonne Université, Hôpital Pitié-Salpêtrière, Univ. Illinois Urbana-Champaign
 - Duration: 3 years (2014-2017). Role Collaborator. Personal contribution: WSI analysis using DL
- **NIH (Nat. Institute Health, US) - Advanced Ultrasonic Evaluation of Sentinel Lymph Nodes**
 - Consortium: Riverside Research, New York NY USA, Hawaii University, Honolulu, USA, Kuakini Medical Center, Honolulu, USA, Duration 5 years (2011-2016);
 - Role: collaborator. Personal contribution: WSI analysis for the classification of metastatic regions in lymph nodes, in correlation with ultrasound modalities.
- **IAMS: Integrated autonomous microscopy syst. for imaging anatomies complex 3D cell**
 - Granted by **A*STAR/JCO** Joint Council Office SERC-BMRC/A*STAR Singapore;
 - Involving IPAL and four A*STAR institutes (the Institute for Infocomm Research I2R/SERC, the Bioinformatic Institute BII/BMRC, the Institute of Molecular and Cell Biology IMCB/BMRC and the Institute of Medical Biology IMB/BMRC);
 - Duration: 3 years (2013 – 2016). Role: PI. Personal contribution: microscopic images analysis.
- **FlexMIm: Collaborative Digital Pathology**
 - Funded by the Consolidated Interministerial Fund, French Ministry of Industry (MINEFE);
 - Project involving Orange Healthcare, Tribvn (SME), Pertimm (SME), the University Pierre and Marie Curie, the Univ. Paris Diderot and the Assistance publique – Hôpitaux de Paris (AP-HP), the public hospital system of the city of Paris and its suburbs;
 - Duration: 3,5 years (2013-2016), Grant: 1.679.000 €, Pers. role: **Principal Investigator**.
- **MICO: COgnitive Microscope: Cognition-driven visual explorer in histopathology**
 - Granted by the Techn. for Health program (TecSan - Technologies pour la Santé) of the **French National Research Agency** (ANR - Agence Nationale de la Recherche);
 - Partners: LIP6 -UPMC, TRIBVN, THALES-TCF, Pitié-Salpêtrière Hospital, AGFA Healthcare;
 - Duration: 3,5 years (01/2011-07/2014). Grant: 1.160.000 €.
 - Personal role: **Project Director & Principal Investigator**
- **Intel. Vision Quantitative Microscopy Neural Stem Cells Progenitor Growth Differentiation**
 - Project funded by **A*STAR/JCO** (Joint Council Office SERC-BMRC/A*STAR) Singapore Involving three A*STAR institutes (Institute for Infocomm Research I2R/SERC, Bioinformatics Inst. BII/BMRC and the Inst. of Medical Biology IMB/BMRC);
 - Duration: 3,5 years (12/2009 – 05/2013). Grant: 741.000 S\$. Personal role: PI.
- **C0604 - EURO-TELEPATH - “Telepathology Network in Europe”**
 - **COST (European Cooperation in Science and Technology)** Action.
 - Partners: Spain, France, Germany, Grece, Italy, Switzerland, Croatia, Finland, Lithuania, Nederland, Norway, Poland, Portugal, United Kingdom;
 - Duration: 4 years (2007-2011). Personal role: expert in semantic imaging
- **MMedWeb (Multimedia Medical Conceptual Web for Intelligent Information Access)**
 - **A*STAR – SERC** grant - Science & Engineering Research Council, Agency for Science, Technology and Research – Singapore.
 - Partners: CNRS, NUS, I2R/A*STAR, Nat. Univ. Hospital (NUH), National Healthcare Group, Singapore General Hospital (SGH);
 - Duration: 3 years (2007-2010). Grant: 530.840 S\$. Personal role: **Principal Investigator**.
- **ONCO-MEDIA (Ontology and COntext related MEDical image Distributed Intel. Access)**
 - **6th ICT-ASIA programme**
 - 11 International partners: Singapore (IPAL), France (CREATIS, LIRIS, LIP6, I3S), Switzerland (UNIGE), Taiwan (NTU), Philippine (ATENEO), Japan (CIGG);
 - Duration of the project: 2 years (2006-2007; 2009-2010);
 - Personal role: **Project Director & Principal Investigator**

AWARDS:

- 1st Prize – Poster contest – European Neuro-Ophtalmology Society EUNOS 2024, Rotterdam, Netherlands - Gungor, A., et al., Early Detection of Central Retinal Artery Occlusion within 4.5 Hours of Visual Loss: Deep Learning Method Applied on Fundus Photographs.
- 1st Place Prize - 2019 IEEE Eng. in Medicine and Biology Prize: Irshad, H. et al. (2014). Methods for Nuclei Detection, Segmentation, and Classification in Digital Histopathology: A Review - Current Status & Future

Potential. *IEEE Reviews on Biomedical Engineering*.

- Best Paper Award finalist - Zemouri, R., Racocceanu, D., Zerhouni, N. (2001). A Petri nets graphic method of reduction using birth-death processes, *IEEE International Conference on Robotics & Automation - IEEE ICRA'2001*, pp. 46-51, Seoul, Korea.

PATENTS:

- Virtual IHC “virtual staining” (paired images) — *Device and method for generating n virtual IHC stain images from one H&E image* — EP 24 305 221.4 (filed 9 Feb 2024), PCT/EP2025/053151 (filed 6 Feb 2025). Institutional co-owner (ICM, AP-HP, CNRS, INSERM, Sorbonne Univ., Univ. Paris Cité); Co-inventor (Racocceanu). PCT publication WO2025/168731 (18 Aug. 2025) Co-inventor : Daniel Racocceanu
- Virtual IHC “virtual staining” (unpaired images) — *Device and method for generating n virtual IHC stain images from one H&E image (unpaired)* — EP 24 305 224.8 (filed 9 Feb 2024), PCT/EP2025/053153 (filed 6 Feb 2025). Institutional co-owner; Co-inventor : Daniel Racocceanu
- 3D cell/particle tracking (microscopy) — *Method and system for tracking motion of microscopic objects within a 3D volume* — US 2014/0192178 A1 (pub. 10 Jul 2014, filed 13 Aug 2012). Holder: A*STAR (Singapore). Co-inventor: Daniel Racocceanu.
- Dynamic real-time monitoring system — INPI (FR) 0858450 (filed 10 Dec 2008). Holder: Em@systec (spin-off UFC). Co-inventor: Daniel Racocceanu.
- Software copyrights (France, CNRS/APP): *Mitosis Detector* (DL 05963-01, 2013); *PDFibAtI@s* (DL 04137-01, 2011); *HISTOGRAD* (DL 2944-01; APP IDDN ... 14 May 2010). Author/co-author; rights held by CNRS/UMI.

CERTIFICATES:

- June 2014, Certif. in Advanced English (CAE), Cambridge English Language Assessment.